Huadong Pump Co., Ltd. (teaching example)

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OPERATION AND MAINTENANCE MANUAL OM-P-205

Centrifugal pump XB80-200 (plant tag P-205)

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Applies to: Pump XB80-200, plant tag P-205

Teaching document of the ML in Industry 2026 course. Not a real manual.

1 Safety

1.1 Read this manual before installation, start-up and maintenance of the pump.

1.2 Only trained personnel may operate and service the pump.

1.3 Stop the pump and lock out the motor before any maintenance work.

2 Description

2.1 The XB80-200 is a single-stage end-suction centrifugal pump with a mechanical seal and oil-lubricated rolling bearings.

2.2 The rated data of the pump are given in Table 1.

Table 1 — Rated data

Parameter	Unit	Value
Flow	m³/h	50
Head	m	80
Speed	r/min	2900
Motor power	kW	37
Mass	kg	520

3 Installation

3.1 Install the pump on a rigid concrete foundation and grout the base plate.

3.2 After installation align the pump and the motor shafts. The radial offset must not exceed 0.08 mm.

3.3 Support the suction and discharge pipes so that they do not load the pump casing.

4 Start-up

4.1 Fill the pump and the suction pipe with liquid before start-up.

4.2 Never run the pump dry, even for a short time.

4.3 Check the direction of rotation by a short start of the motor.

5 Condition monitoring

5.1 Measure the vibration velocity (RMS, 10 to 1000 Hz) on both bearing housings once a week.

5.2 The vibration limits of this pump are given in Table 2. They apply to the pump on a rigid foundation.

5.3 If the alarm limit is exceeded, find the cause within 7 days (see section 7).

5.4 If the trip limit is exceeded, stop the pump at once.

5.5 The temperature of the bearing housing must not exceed 95 °C.

Table 2 — Vibration limits (velocity, mm/s RMS)

Zone	Velocity, mm/s	Action
Normal operation	up to 7.1	none
Alarm	above 7.1	find the cause within 7 days
Trip	above 11.2	stop the pump

6 Maintenance

6.1 The bearings are oil-lubricated. Change the oil every 3000 operating hours.

6.2 The oil volume of the bearing housing is 0.6 L.

6.3 Replace the mechanical seal when the leakage exceeds 5 drops per minute.

6.4 Replace the rolling bearings after 20 000 operating hours.

7 Troubleshooting

7.1 Typical causes of high vibration and their remedies are given in Table 3.

Table 3 — Troubleshooting

Symptom	Possible cause	Remedy
High vibration	misalignment	align the shafts again
High vibration	worn bearings	replace the bearings
High vibration	cavitation	check the suction pressure and the filter
Hot bearing	too much or too little oil	set the correct oil level